Chapter 10: Least squres estimation with endogenous regressors

Financial Econometrics

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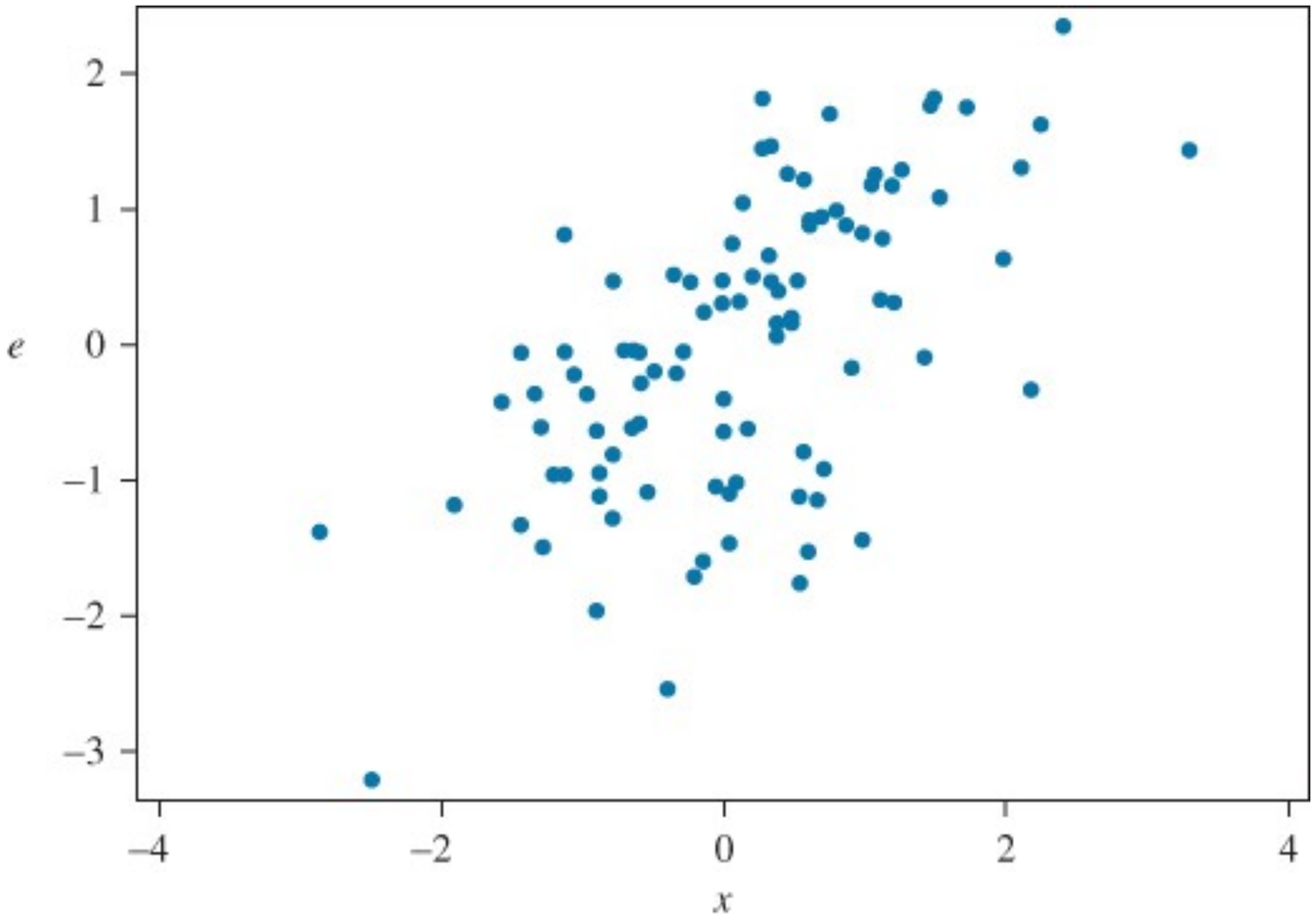
Introduction

- In most data coming from natural, non-controlled phenomena, both the dependent and independent variables are random: (x, y) are random
- In many cases, some of the independent variables are also correlated with the error term in a regression model, i.e., $cov(x, e) \neq 0$, which makes the OLS method inappropriate. (Think about it: How do you check this?)

Fig 10.1 (a) Correlated *x* and *e*

Figure 10.1 (a)



Roadmap

10.1 Least squares estimation with endogenous regressors

10.2 Endogeneity problem: $cov(x, e) \neq 0$

10.3 Instrumentalvariables (IVs): z_l for $l = 1, \ldots, L$

- (1) z_l does not direct affect y; (2) $cov(z_l, e) = 0$; (3) $cov(x, z_l) \neq 0$
- Surplus moment conditions: $L \geq B$
- Method of moments (MM)
- 2-stage least squares estimation (2SLS)
- Generalized method of moments estimation (GMM) for hetereoskedasticity

10.4 Specification tests

- 10.4.1-2 Hausman test for endogeneity

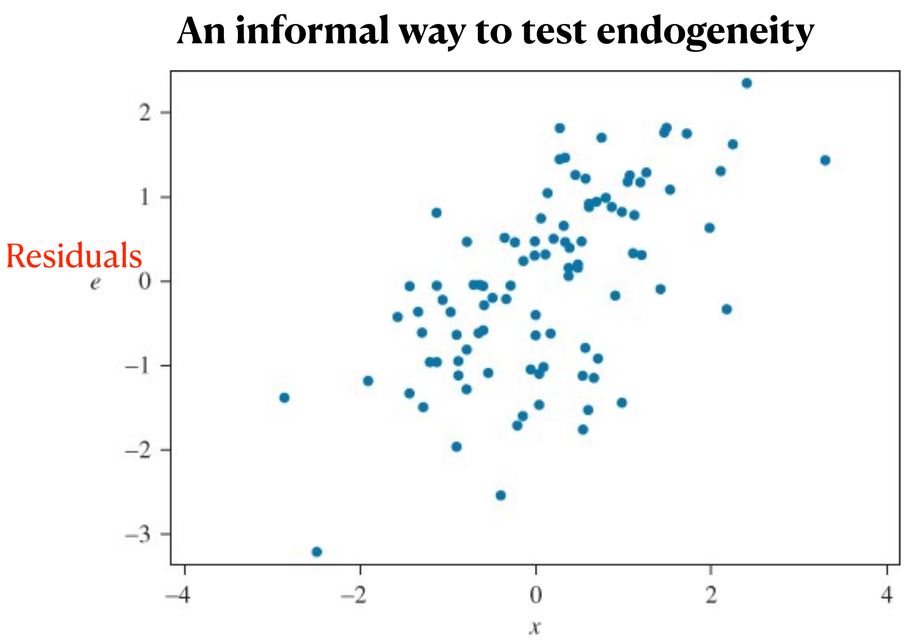
  $H_0 : cov(x, e) = 0$

- 10.3.9 Weak instruments tests

H_0: $z_1, \ldots, z_L$ are weak IVs

- 10.4.3 Overidentifying test (Sargen test)

H_0: IVs are valid



10.1 Least squares estimation with endogenous regressors

The Simple linear regression under random sampling

- RS1: The observable variables y and x are related by $y_i = \beta_1 + \beta_2 x_2 + e_i$, $i = 1, \ldots, N$, where β_1 and β_2 are unknown population parameters and e_i is a random error term.
- RS2: The data pairs (y_i, x_i) are statistically independent of all other data pairs and have the same joint distribution $f(y_i, x_i)$. The are independent and identically distributed (iid).
- RS3: $E(e_i \mid x_i) = 0$ for $i = 1, \ldots, N$; x is contemporaneously, and strictly, exogenous.
- RS4: The random error has constant conditional variance, $var(e_i \mid x_i) = \sigma^2$.
- RS5: x_i takes at least two different values.
- RS6: $e_i \sim N(0, \sigma^2)$.

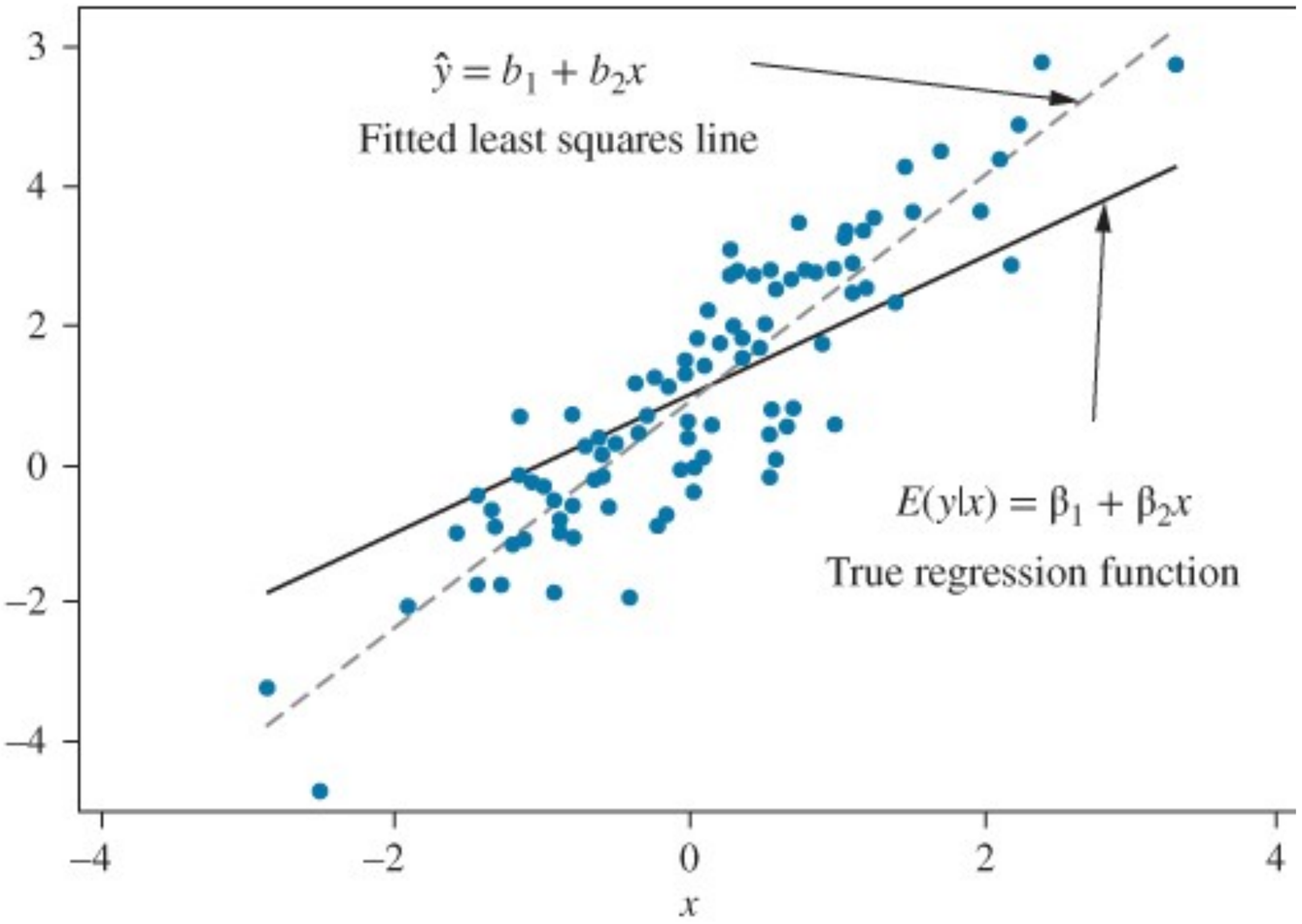
- We can replace RS3 by RS3*: $\mathsf{E}(e_i) = 0$ and $\text{cov}(x_i, e_i) = 0$.
- Recall
  - An estimator $\hat{\theta}_N$ is unbaised, if $\mathsf{E}[\hat{\theta}] = \theta$.
  - An estimator $\hat{\theta}_N$ is consistent, if $\hat{\theta}_n \xrightarrow{p} \theta$
- What if $cov(x_i, e_i) \neq 0$?

10.1.1. Large sample properties of the OLS estimator

Assume RS1, RS2, RS4, RS5 hold

	$cov(x_i, e_i) = 0$	$cov(x_i, e_i) \neq 0.$
Definition	exogenous regressors	endogenous regressors
Properties of LSE	biased	biased
	consistent	inconsistent
Estimation	MM	MM
Note	MM = LSE	MM$\neq$ LSE

Fig 10.1 (b) Plot of data, true, and fitted regression functions



When $cov(x_i, e_i) = 0$, LSE consistent!

The regression model:

$$y_i = \beta_1 + \beta_2 x_i + e_i \,.$$

Assume $E(e_i) = 0$, then $E(y_i) = \beta_1 + \beta_2 E(x_i)$:

$$y_i - E(y_i) = \beta_2[x_i - E(x_i)] + e_i \,.$$

Multiply the above equation with $(x_i - E(x_i))$ and take an expectation:

$$[x_i - E(x_i)][y_i - E(y_i)] = \beta_2[x_i - E(x_i)]^2 + [x_i - E(x_i)]e_i \,,$$

$$E\{[x_i - E(x_i)][y_i - E(y_i)]\} = \beta_2 E\,\{[x_i - E(x_i)]^2\} + E\,\{\,[x_i - E(x_i)]e_i\,\}.$$

$$\Longrightarrow cov(x_i, y_i) = \beta_2 var(x_i) + cov(x_i, e_i)$$

Solve for β_2,

$$\beta_2 = \frac{cov(x_i, y_i)}{var(x_i)} - \frac{cov(x_i, e_i)}{var(x_i)}.$$

Assuming $cov(x_i, e_i) = 0$, we have

$$\beta_2 = \frac{cov(x_i, y_i)}{var(x_i)}.$$

1. LSE is biased: $\mathsf{E}(b_2) = \mathsf{E}\left[\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}\right] \overset{?}{\neq} \beta_2$

2. The LSE is consistent:

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\hat{cov}(x, y)}{\hat{var}(x)} \to \frac{cov(x, y)}{var(x)} = \beta_2.$$

This means that the least squares estimator is consistent.

When $cov(x_i, e_i) \neq 0$, LSE inconsistent!

If $cov(x_i, e_i) \neq 0$,

$$\beta_2 = \frac{cov(x_i, y_i)}{var(x_i)} - \frac{cov(x_i, e_i)}{var(x_i))} .$$

Then, the least squares estimator b_2 converges to

$$b_2 \xrightarrow{p} \frac{cov(x, y)}{var(x)} = \beta_2 + \frac{cov(x, e)}{var(x)} \neq \beta_2 .$$

b_2 is an inconsistent estimator of β_2 !

The amount of bias that exists even asymptotically, even when samples can be assumed to be larger!

10.2 Cases with Which x and e are Contemporaneously Correlated

When an explanatory variable and the error term are contemporaneous correlated, i.e. $cov(x_i, e_i) \neq 0$, the explanatory variable is said to be endogenous.

This term comes from simultaneous equation models (ch11).

It means "determined within the system".

Using this terminology when $cov(x_i, e_i) \neq 0$, one is said to have an *endogeneity problem*.

10.2.1 Measurement error

Consider measurement error:

$$x = x^* + u,$$

where $x^*$is the real explanatory variable and u is a random error.

The regression model is:

$$\begin{aligned} y &= \beta_1 + \beta_2 x^* + e \\ &= \beta_1 + \beta_2 (x - u) + e \\ &= \beta_1 + \beta_2 x + (e - \beta_2 u) \end{aligned}$$

Then, $cov(x, e - \beta_2 u) = cov(x^* + u, e - \beta_2 u) = cov(u, -\beta_2 u) = -\beta_2 var(u) \neq 0.$

10.2.2 Simultaneous equation bias

- Supply-demand model:

$$Q(\xi) : \text{quantity},$$

$$P(\xi) : \text{price},$$

$$\xi : \xi \text{ is some common factor (or the system)},$$

$Q(\xi) = \beta_1 + \beta_2 P(\xi) + e$, where e is a random error.

There is a feedback relationship between P_i and Q_i.

It can be shown that $cov(P, e) \neq 0$ (Proof skipped)

10.2.3 Lagged dependent variable models with serial correlation

Consider a lagged dependent variabe:

$$y_t = \beta_1 + \beta_2 y_{t-1} + \beta_3 x_t + e_t \,.$$

Assume e_t follow the AR(1) process,

$$e_t = \alpha e_{t-1} + \varepsilon_t \,,$$

for some v and white noise ε_t .

Then $cov(y_{t-1}, e_t) \neq 0$:

$$\begin{aligned} cov(y_{t-1}, e_t) &= cov(y_{t-1}, \alpha e_{t-1} + \varepsilon_t) \\ &= cov(\beta_1 + \beta_2 y_{t-2} + \beta_3 x_{t-1} + e_{t-1}, \alpha e_{t-1} + \varepsilon_t) \\ &= cov(e_{t-1}, \alpha e_{t-1}) = \alpha var(e_{t-a}) \neq 0 \,. \end{aligned}$$

Thus, the lagged variable y_{t-1} is a random regressor.

10.2.4 Omitted variables

Recall a general form of a model:

$$y = m(\theta, x_1, \ldots, x_p) + e \ ,$$

one needs to decide the functional form and the explanatory variables in $m(\theta, x)$.

Suppose the ideal and real model is:

$$y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + v, \ v \sim N(0, \sigma^2) \ .$$

Assume x_3 is the omitted variable and $cov(x_2, x_3) \neq 0$.

Consider a working model:

$$y = \beta_1 + \beta_2 x_2 + e,\ e \sim \text{error}\,.$$

We would like to show $cov(x_2, e) \neq 0$ and conclude x_2 is an endogenous regressor.

$$\begin{aligned} cov(x_2, e) &= cov(x_2, \beta_3 x_3 + v) \\ &= \beta_3 cov(x_2, x_3) + cov(x_2, v) \\ &= \beta_3 cov(x_2, x_3) \neq 0\,. \end{aligned}$$

Example 10.1

EXAMPLE 10.1 | Least Squares Estimation of a Wage Equation

We will use the data on married women in the data file *mroz* to estimate the wage model in (10.6). Using the $N = 428$ women in the sample who are in the labor force, the least squares estimates and their standard errors are

$$\ln(WAGE) = -0.5220 + 0.1075 \times EDUC + 0.0416 \times EXPER - 0.0008 \times EXPER^2$$

$$\text{(se)} \quad (0.1986) \quad (0.0141) \quad (0.0132) \quad (0.0004)$$

We estimate that an additional year of education increases wages by approximately 10.75%, holding everything else constant. If ability has a positive effect on wages, then this estimate is overstated, as the contribution of ability is attributed to the education variable.

The social and policy importance of the estimate 0.1075 can hardly be exaggerated. Countries invest a large portion of tax revenue to improve education. Why? Spending on education is an investment, and like any other investment investors (taxpaying citizens) expect a rate of return that is competitive with rates of returns for alternative projects. Based on the estimated equation above, additional years of schooling are estimated to increase wages by 10.75%, holding other factors fixed, meaning that individuals are

10.3 Estimators Based on the Method of Moments

10.3 Estimators Based on the Method of Moments

When all the usual assumptions of the linear model hold, the method of moments leads to the least squares estimator.

If x_i is random and correlated with the error term, the method of moments leads to an alternative, called, instrumental variables estimation, or two-stage least squares estimation, that will work in large samples.

Overview of Estimation methods

- Least squares estimators (LSE, or OLS): Ch 2, Ch 5
- Generalized least squares (GLS): Ch 8
- The method of moments (MM): Ch 10
- The maximum likelihood estimators (MLE): Ch 16
- Bayesian inference
- Indeed, there are more…

Solutions to endogenenity problems

1. Instrumental variable (IV)
2. Using method of moments as an alternative estimator procedure for parameters
3. Difficulty in using IV
4. Two-Stage Least Squares Estimation

10.3.1. Method of moments estimation (MM) of a population mean and variance

The kth moment of Y is

$$\mu_k = E(Y^k) \, .$$

It can be estimated consistently using the sample (of size N) analog:

$$\hat{\mu}_k = \frac{1}{N} \sum y_i^k \, .$$

The law of Large Numbers (LLN)

- If $X_1, \ldots, X_N$ is a random sample from a population, and if $E(X_i) = \mu < \infty$ and $\text{var}(X_i) = \sigma^2 < \infty$, then the sample mean $\bar{X} = \sum X_i/N$ converges (in probability) to the expected value (population mean) μ as the sample size N increases.
  - We denote the above by: $\bar{X} \xrightarrow{p} \mu$.
- In this case, $\bar{X}$ is said to be a consistent estimator of μ.
- In most situations, sample moments are consistent estimators of population moments.

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Recall: Convergence of a sequence a_n

We say a_n approaches a as n approaches infinity, denoted as

$$\lim_{n\to\infty} a_n = a,$$

or,

$$a_n \to a \text{ as } n \to \infty,$$

if for any δ, you can find always find an $N = N(\delta)$, such that

$$|a_n - a| < \delta, \text{ if } n > N.$$

Plain language

Math Notations

Definition

Definition: Convergence in probability

We say that a sequence of random variables $\{Z_n\}$ converge to α, denoted as

$$Z_n \to \alpha \text{ in probability}$$

Or

$$Z_n \xrightarrow{p} \alpha$$

if for some $\varepsilon > 0$, $P(|Z_n - \alpha| > \varepsilon) \to 0$ as $n \to \infty$.

S.O.P. of MM

Suppose you have m parameters $\theta_1, \ldots, \theta_m$:

1. Express m moments:

$$\mu_k = f_k(\theta_1, \ldots, \theta_m) \text{ for } k = 1, \ldots, m\,.$$

2. Represent these parameters using μ_k:

$$\theta_k = g_k(\mu_1, \ldots, \mu_m) \text{ for } k = 1, \ldots, m\,.$$

3. Use $\hat{\mu}_k$ to estimate μ_k :

$$\hat{\theta}_k = g_k(\hat{\mu}_1, \ldots, \hat{\mu}_m) \text{ for } k = 1, \ldots, m\,.$$

MM1 : $Y \sim N(\mu, \sigma^2)$

1. Express the μ_1 and μ_2 by μ and σ^2 :

$$\mu_1 = E(Y) = \mu \ ,$$

$$\mu_2 = E(Y^2) = \sigma^2 + \mu^2$$

2. Represent the two parameters as a function of μ_k :

$$\mu = \mu_1$$

$$\sigma^2 = \mu_2 - \mu_1^2$$

3. Estimate parameters by plugging $\hat{\mu}_k$:

$$\hat{\mu} = \hat{\mu}_1 = \sum y_i / N = \bar{y}$$

$$\tilde{\sigma}^2 = \hat{\mu}_2 - \hat{\mu}_1^2 = \frac{1}{N} \sum y_i^2 - \bar{y}^2 = \frac{\sum (y_i - \bar{y})^2}{N}.$$

$\tilde{\sigma}^2$is biased but consistent! $S^2 = \dfrac{\sum (y_i - \bar{y})^2}{N-1}$is unbiased.

MM 2: the simple regression model (10.3.2)

In the simple regression model, $y = \beta_1 + \beta_2 x + e$, we have two parameters : β_1 and β_2 .

Assume $cov(x, e) = 0$.

1. Two moments conditions: $E(e) = 0$ and $E(ex) = 0$ Because $cov(x, e) = E(xe) - E(x)E(e) = E(xe) = 0$

$$E(e) = 0 \Rightarrow E(y - \beta_1 - \beta_2 x) = 0$$

$$E(xe) = 0 \Rightarrow E[x(y - \beta_1 - \beta_2 x)] = 0$$

Simplified results:

$$E(y) = \beta_1 + \beta_2 E(x)$$

$$E(xy) = \beta_1 E(x) + \beta_2 E(x^2)$$

2. Represent parameters using moment conditions:

$$\beta_2 = \frac{E(xy) - E(x)E(y)}{E(x^2) - (E(x))^2} = \frac{E\left[(x - E(x))(y - E(y))\right]}{E(x - E(x))^2},$$

$$\beta_1 = E(y) - \beta_2 E(x) .$$

3. Plugging sample moments:

$$\hat{\beta}_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2},$$

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}.$$

$\hat{\beta}_1$ and $\hat{\beta}_2$ are exactly the same as the LSE b_1 and b_2!

Note: MM estimators are not necessarily to be equal to the LS estimators!

10.3.3. IVs estimation in the simple regression model ($B = L = 1$)

Suppose that there is another variable, z_i , such that

1. z_i does not have a direct effect on y_i, and thus it does not belong on the right-hand side of the model as an explanatory variable.

2. z_i is not correlated with the regression error$e$. (i.e., z_i is exogenous, $corr(z_i, e_i) = 0$)

3. z_i is strongly (or at least not weekly) correlated with x_i, the endogenous variable (i.e., $corr(z_i, x_i) \neq 0$).

A variable z_i with these properties is called an instrumental variable (IV).

B bad explanatory variables, and L lucy instrumental variables.

MM 3: IV in the simple regression model

1. Two moment conditions (using IV, z_i):

$$E(e) = 0 \Rightarrow E[(y - \beta_1 - \beta_2 x)] = 0.$$
$$E(ze) = 0 \Rightarrow E[z(y - \beta_1 - \beta_2 x)] = 0.$$

Simplified results:

$$E(y) = \beta_1 + \beta_2 E(x)$$
$$E(yz) = \beta_1 E(z) + \beta_2 E(xz)\,.$$

2. Express parameters using moments:

$$\beta_2 = \frac{E(yx) - E(y)E(z)}{E(xz) - E(x)E(z)} = \frac{E((y - E(y))(z - E(z)))}{E((x - E(x))(z - E(z)))}$$
$$\beta_1 = E(y) - \beta_1 E(x)\,.$$

3. Plug in sample moments to obtain the MM estimators:

$$\hat{\beta}_2 = \frac{\sum (z_i - \bar{z})(y_i - \bar{y})}{\sum (z_i - \bar{z})(x_i - \bar{x})},$$
$$\hat{\beta}_1 = \hat{y} - \hat{\beta}_2 \bar{x}\,.$$

The solutions are called the instrumental variable (IV) estimators, and they are consistent!

Check page 493 in the textbook for details

In large samples, we have

$$\hat{\beta}_2 \xrightarrow{d} N\left(\beta_2, \hat{var}(\hat{\beta}_2)\right),$$

where

$$\hat{var}(\hat{\beta}_2) = \frac{\hat{\sigma}_{IV}^2 \sum (z_i - \bar{z})^2}{\left[\sum (z_i - \bar{z})(x_i - \bar{x})\right]^2},$$

and

$$\hat{\sigma}_{IV}^2 = \frac{\sum (y_i - \hat{\beta}_1 - \hat{\beta}_2)^2}{N - 2}.$$

10.3.4 The importance of using strong instruments

Suppose $cov(x_i, e_i) = 0$ so that both OLS and IV estimators are consistent. Then, it can be shown that

$$\frac{\hat{var}(\hat{\beta}_2)}{\hat{var}(b_2)} = \frac{\dfrac{\hat{\sigma}^2_{IV} \sum (z_i - \bar{z})^2}{\left[\sum (z_i - \bar{z})(x_i - \bar{x})\right]^2}}{\hat{\sigma}^2 / \sum (x_i - \bar{x})^2} = \frac{\hat{\sigma}^2_{IV} / \hat{\sigma}^2}{r^2_{zx}} \approx \frac{1}{r^2_{zx}}.$$

Because $r^2_{zx} < 1$, the IV estimator is less efficient than the OLS estimator.

You must search a strong IV that is highly correlated with the endogenous x.

10.3.5 Proving the consistency of the IV estimator

Consistency! Because the sample covariance converges to the true covariance in large sampel, we have

$$\hat{\beta}_2 = \frac{\sum (z_i - \bar{z})(y_i - \bar{y})}{\sum (z_i - \bar{z})(x_i - \bar{x})} \xrightarrow{p} \frac{cov(z, y)}{cov(z, x)} .$$

Recall that we need

1. $cov(z_i, x_i) \neq 0$;
2. $cov(z_i, e_i) = 0$.

Example 10.2

- corr(EDU,Ability) >0
- Is MOTHEREDU a good IV?
  1. MOTHEREDU not effect on WAGE
  2. Corr(MOTHEREDU, ABILITY)=0
  3. Corr(METHEREDU, EDUC)=0.3870

PLAN to STOP HERE (20250421)!!!

EXAMPLE 10.2 | IV Estimation of a Simple Wage Equation

To illustrate the instrumental variables estimation method in a simple regression consider a simplified version of the model used in Example 10.1, $\ln(WAGE) = \beta_1 + \beta_2 EDUC + e$. Using the data file *mroz* on $N = 428$ married women, the OLS estimates are

$$\widehat{\ln(WAGE)} = -0.1852 + 0.1086EDUC$$
$$\text{(se)} \qquad (0.1852) \quad (0.0144)$$

The estimated rate of return to education is approximately 10.86%, and $t = 7.55$ indicates that the estimated coefficient is significantly different from zero at even the 1% level of significance. If *EDUC* is endogenous, and correlated with the random error e, then OLS estimation may lead to incorrect inferences. We anticipate that *EDUC* is positively correlated with the omitted variable "ability," meaning that the estimated rate of return 10.86% may overstate the true value.

What might we use as an instrumental variable? One proposal is to use mother's years of education, *MOTHEREDUC*, as an instrument. Does this qualify? In Section 10.3.3, we listed three criteria for an instrumental variable. First, does this variable have a direct effect on the dependent variable? Does it belong in the equation? Mother's education should not play any direct role in the determination of a daughter's wage, so this seems fine. Second, the instrument should not be contemporaneously correlated with the random error, e. Is a mother's education correlated with the omitted variable, her daughter's ability? This is more difficult. Ability includes many attributes, including intelligence, creativity, perseverance, and industriousness to name a few. Some portion of these character traits may be passed into our genetic makeup from our parents. We dodge the scientific debate on this issue and assume that a mother's years of education are uncorrelated with her daughter's ability. Third, the instrument should be highly correlated with the endogenous variable. This we can check! For the 428 women in the sample the correlation between mother's education and daughter's education is 0.3870. This is not very large, but it is not very small either.

The instrumental variables estimates are

$$\widehat{\ln(WAGE)} = 0.7022 + 0.0385EDUC$$
$$\text{(se)} \qquad (0.4851) \quad (0.0382)$$

The IV estimate of the rate of return to education is 3.85%, one-third of the OLS estimate. The standard error is about 2.65 times larger than the OLS standard error, which is very close to what we reasoned that the ratio might be when both estimators are consistent,

$$\text{se}\left(\hat{\beta}_2\right) \Big/ \text{se}(b_2) = 0.0382/0.0144 = 2.65 \simeq 1/r_{zx}$$
$$= 1/0.3807 = 2.58$$

10.3.6. IV estimation using two-stage least squares (2SLS) ($B = L = 1$)

1. The first stage equation is

$$x = \gamma_1 + \theta_1 z + v,$$

where γ_1 is an intercept parameter, θ_1 is a slope parameter, and ν is an error.

Estimate the first-stage equation by OLS and obtain the fitted value,

$$\hat{x} = \hat{\gamma}_1 + \hat{\theta}_1 z\,.$$

2. Replace the endogenous variable x in the simple regression $y = \beta_1 + \beta_2 x + e$ with $\hat{x} = \hat{\gamma} + \hat{\theta}_1 z$ and then apply OLS to

$$y = \beta_1 + \beta_2 \hat{x} + e$$

The OLS estimates of β_1 and β_2 from the second-stage regression are identically equal to the IV estimates $\hat{\beta}_1$ and $\hat{\beta}_2$.

But, OLS produces wrong variance estimate!

Example 10.3

EXAMPLE 10.3 | 2SLS Estimation of a Simple Wage Equation

To illustrate the two-stage least squares equivalent of instrumental variables estimation, we estimate the first-stage equation, a regression of the endogenous variable *EDUC* on the instrumental variable *MOTHEREDUC*

$$\widehat{EDUC} = 10.1145 + \boxed{0.2674}MOTHEREDUC$$
$$\text{(se)} \quad (0.3109) \quad \boxed{(0.0309)}$$

In order for *MOTHEREDUC* to be a strong instrumental variable it must be strongly correlated with *EDUC*. Another way to say this is that *MOTHEREDUC* should be strongly significant in this first-stage equation, and it is. The t-value is 8.66, so the coefficient is significantly different from zero at the 1% level. We will say much more about this approach in Section 10.3.9.

In the second-stage equation, we regress ln(*WAGE*) on the fitted value from the first-stage equation,

$$\widehat{\ln(WAGE)} = 0.7021 + \boxed{0.0385}\widehat{EDUC}$$
$$\text{(incorrect se)} \quad (0.5021) \quad \boxed{(0.0396)}$$

Wrong!

The coefficient estimates are the same as in Example 10.2, but note that the standard errors produced by this second OLS estimation are not the same as in Example 10.2. They are **incorrect** because they use $\hat{\sigma}^2_{WRONG}$.

10.3.7 Using surplus moment conditions

Sometimes, we have more IV than necessary.

Suppose we have two good instruments, z_1 and z_2 that satisfy conditions IV1-IV3. Then, we have

$$E[z_2 e] = E[z_2(y - \beta_1 - \beta_2)] = 0.$$

There are now three sample moment conditions:

$$\frac{1}{N}\sum (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) = \hat{m}_1 = 0\,,$$
$$\frac{1}{N}\sum z_{i1}(y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) = \hat{m}_2 = 0\,,$$
$$\frac{1}{N}\sum z_{i2}(y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i) = \hat{m}_3 = 0.$$

Surplus moment conditions: Difficulties in MM

Difficulties in MM

- We have three equations with only two unknown parameters.
- There are no solutions satisfying all three equations.

A solution using 2SLS to include all IVs

- It can be proved that the best way of combining instruments is using the 2SLS idea.
- If we have more than two IV, we apply the same strategy.

Example 10.4

EXAMPLE 10.4 | Using Surplus Instruments in the Simple Wage Equation

Father's education is also a potential instrument for daughter's education. Using the 428 observations in the data file *mroz*, the correlation between *FATHEREDUC* and *EDUC* is 0.4154. The first-stage equation is

$$\boxed{EDUC = \gamma_1 + \theta_1 MOTHEREDUC + \theta_2 FATHEREDUC + v}$$

The OLS estimated first-stage equation is

$$\boxed{\begin{aligned} \widehat{EDUC} &= 9.4801 + 0.1564 MOTHEREDUC \\ (\text{se}) \quad & (0.3211) \quad (0.0358) \\ &+ 0.1881 FATHEREDUC \\ & (0.0336) \end{aligned}}$$

The t-statistics for the coefficients of *MOTHEREDUC* and *FATHEREDUC* are 4.37 and 5.59, respectively, and are significant at the 1% level. The test of the joint significance of the two IV is even more important than their individual significance. The F-statistic for the null hypothesis $H_0: \theta_1 = 0, \ \theta_2 = 0$ is 55.83, which is very significant, and we can conclude that at least one of the two IV coefficients is not zero based on this joint test. The importance of the F-test is discussed in Section 10.3.9.

In the second-stage equation, we replace *EDUC* by $\widehat{EDUC}$ and apply least squares to obtain the IV/2SLS estimates

$$\begin{aligned} \widehat{\ln(WAGE)} &= 0.5510 + 0.0505 \widehat{EDUC} \\ (\text{incorrect se}) \quad & (0.4257) \quad (0.0335) \end{aligned}$$

The coefficient estimates are the correct IV estimates, but the standard errors reported are incorrect. Using proper IV software yields

$$\begin{aligned} \widehat{\ln(WAGE)} &= 0.5510 + \boxed{0.0505 \widehat{EDUC}} \\ (\text{se}) \quad & (0.4086) \quad \boxed{(0.0322)} \end{aligned}$$

10.3.8 IV estimation in the multiple regression model

B=1, bad endogenous variables , L: arbitrary and luck IVs, G: good, good exogenous varaibles

1. The first stage regression:$x_K = \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_{K-1} x_{K-1} + \theta_1 z_1 + \cdots + \theta_L z_L + v_K$, where ν_K is a random error that is uncorrelated with all the right-hand side variables. Use OLS to estimate γ and θ to obtain the fitted value of x_K:

$$\hat{x}_K = \hat{\gamma}_1 + \hat{\gamma}_2 x_2 + \cdots + \hat{\gamma}_{K-1} x_{K-1} + \hat{\theta}_1 z_1 + \cdots + \hat{\theta}_L z_L \,.$$

2. The second stage regression:

$$y = \beta_1 + \beta_2 x_2 + \cdots + \beta_K \hat{x}_K + e^*,$$

where e^* is a random term. OLS from this equation, $\hat{\beta}_1, \ldots, \hat{\beta}_K$, are the instrumental variables(IV) estimator, and are also known as two-stage least squares (2SLS) estimators, or, IV/2SLS estimators.

Example 10.5

EXAMPLE 10.5 | IV/2SLS Estimation in the Wage Equation

In addition to education a worker's experience is also important in determining their wage. Because additional years of experience have a declining marginal effect on wage use the quadratic model

$$\ln(WAGE) = \beta_1 + \beta_2 EXPER + \beta_3 EXPER^2 + \beta_4 EDUC + e$$

where *EXPER* is years of experience. This is the same specification as in Example 10.1. We assume that *EXPER* is an **exogenous** variable that is uncorrelated with the worker's ability and therefore uncorrelated with the random error e. Two instrumental variables for years of education, *EDUC*, are mother's and father's years of education, *MOTHEREDUC* and *FATHEREDUC*, introduced in the previous examples. The first-stage equation is

$$EDUC = \gamma_1 + \gamma_2 EXPER + \gamma_3 EXPER^2 + \theta_1 MOTHEREDUC + \theta_2 FATHEREDUC + v$$

Using the 428 observations in the data file *mroz* the estimated first-stage equation is reported in Table 10.1. The IV/2SLS estimates, with correctly computed standard errors, are

$$\widehat{\ln(WAGE)} = \underset{(0.4003)}{0.0481} + \underset{(0.0134)}{0.0442}EXPER - \underset{(0.0004)}{0.0009}EXPER^2 + \underset{(0.0314)}{0.0614}EDUC$$

(se)

Significant!

The estimated return to education is approximately 6.1%, and the estimated coefficient is statistically significant with a $t = 1.96$.

TABLE 10.1 **First-Stage Equation**

Variable	Coefficient	Std. Error	*t*-Statistic	Prob.
C	9.1026	0.4266	21.3396	0.0000
EXPER	0.0452	0.0403	1.1236	0.2618
$EXPER^2$	−0.0010	0.0012	−0.8386	0.4022
MOTHEREDUC	0.1576	0.0359	4.3906	0.0000
FATHEREDUC	0.1895	0.0338	5.6152	0.0000

Example 10.6

In the first-stage equation

- T-test for each IV
- F-test.
- R^2

EXAMPLE 10.6 | Checking Instrument Strength in the Wage Equation

In Example 10.5, there is only one potentially endogenous variable in the wage equation, *EDUC*. The minimum number of instrumental variables is one. Given two instruments, we require that at least one of them be significant in the first-stage equation. The F-test null hypothesis is that both coefficients, θ_1 and θ_2, are zero, and if we reject this null hypothesis we conclude that at least one of them is nonzero. In the first-stage regression in Table 10.1, the estimated coefficient of *MOTHEREDUC* is 0.1576 with a t-value of 4.39, and the estimated coefficient of *FATHEREDUC* is 0.1895 with a t-value of 5.62. The F-statistic value for the null hypothesis that both these coefficients are zero is 55.40, which is significant at the 1% level, but more importantly it is larger than the rule-of-thumb threshold, $F > 10$. In addition to the vitally important F-statistic, the goodness-of-fit measures R^2 and $\bar{R}^2$ are sometimes reported. For the first-stage equation in Table 10.1, these values are $R^2 = 0.1527$ and $\bar{R}^2 = 0.1467$.

Partial Correlation and Partial R^2

In addition to the first-stage F-statistic, R^2 and adjusted-R^2, a partial correlation or partial-R^2 are informative. Applying the partialling-out strategy of the Frisch–Waugh–Lovell Theorem is another way to examine instrument strength. The included exogenous variables in the wage equation are $\mathbf{x}_{exog} = (x_1 = 1, EXPER, EXPER^2)$. Regress *EDUC* on $\mathbf{x}_{exog}$ and obtain the residuals, *REDUC*.

Suppose that we are using the single instrument *MOTHEREDUC*. Regress *MOTHEREDUC* on $\mathbf{x}_{exog}$ and obtain the residuals, *RMOM*. These residual variables have the included exogenous variables partialled-out. That is, we have removed the linear influences of the included exogenous variables from the endogenous variable *EDUC* and the external IV, *MOTHEREDUC*. The correlation between *REDUC* and *RMOM* is called a **partial correlation**, and in this case it is 0.3854. The R^2 from a regression of *REDUC* on *RMOM* is called the partial-R^2, and in this case it is 0.1485. Because we have one endogenous variable and one external IV, the partial-$R^2 = 0.1485$ is the square of the partial correlation, $0.3854^2 = 0.1485$.

If there are more external instruments, the partial-R^2 is the R^2 of the partialled-out endogenous variable on all the partialled-out external IV. Add *FATHEREDUC* as an IV, regress it on $\mathbf{x}_{exog}$ and obtain the residuals, *RDAD*. The partial-R^2 is then the R^2 from the regression of *REDUC* on *RMOM* and *RDAD*. In this case, partial-$R^2 = 0.2076$ and the adjusted partial-$R^2 = 0.2038$.

10.3.10 IV estimation in a general model

The multiple regression model, including all K variables, is

$$y = \beta_1 + \beta_2 x_2 + \cdots + \beta_G x_G + \beta_{G+1} x_{G+1} + \cdots + \beta_{G+B} x_K + e,$$

with G good explanatory variables, B bad explanatory variables, and L lucy instrumental variables.

It is a necessary condition for IV estimation that $L \geq B$.
If $L = B$, there are just enough IVs to carry out IV estimation.

If $L > B$, we have more instruments than necessary for IV estimation, and the model is said to be overidentified.

SOP in 2SLS

1. Consider the B first-stage equations:

$$x_{G+j} = \gamma_{1j} + \gamma_{2j}x_2 + \cdots + \gamma_{Gj}x_G + \theta_1 z_1 + \cdots + \theta_{Lj}z_L + \nu_j, j = 1,\ldots,B\,.$$

   The predicted values are

$$\hat{x}_{G+j} = \hat{\gamma}_{1j} + \hat{\gamma}_{2j}x_2 + \cdots + \hat{\gamma}_{Gj}x_G + \hat{\theta}_1 z_1 + \cdots + \hat{\theta}_{Lj}z_L, j = 1,\ldots,B\,.$$

2. We apply LS to the second-stage equation:

$$y = \beta_1 + \beta_2 x_2 + \cdots + \beta_G x_G + \beta_{G+1}\hat{x}_{G+1} + \ldots + \beta_K \hat{x}_K + e^*,$$

Example: $B = L = 2$

Consider the model with $B = 2$ and $L = 2$:

$$y = \beta_1 + \beta_2 x_2 + \cdots + \beta_G x_G + \beta_{G+1} x_{G+1} + \beta_{G+2} x_{G+2} + e\,.$$

1. The first-stage equations are:

$$x_{G+1} = \gamma_{11} + \gamma_{21} x_2 + \cdots + \gamma_{G1} x_G + \theta_{11} z_1 + \theta_{21} z_2 + \nu_1,$$

$$x_{G+2} = \gamma_{12} + \gamma_{22} x_2 + \cdots + \gamma_{G2} x_G + \theta_{12} z_1 + \theta_{22} z_2 + \nu_2,$$

2. The second-stage equation is to apply least squares to

$$y = \beta_1 + \beta_2 x_2 + \cdots + \beta_G x_G + \beta_{G+1} \hat{x}_{G+1} + \beta_{G+2} \hat{x}_{G+2} + e^*\,.$$

10.3.11 Additional issues when using IV estimation

When testing $H_0 : \beta_k = c$, use of the test statistic,

$$t_{STAT} = \frac{\hat{\beta} - c}{se(\hat{\beta})} \sim t_{(df)},$$

is valid. Tests based on the t distribution tend to work better in samples of data that are not large.

To test a joint hypothesis, $H_0 : \beta_2 = c_2, \beta_3 = c_3,$ the test statistic is

$$\chi^2_{STAT} = \left(\frac{\hat{\beta}_2 - c_2}{se(\hat{\beta}_2)}\right)^2 + \left(\frac{\hat{\beta}_3 - c_3}{se(\hat{\beta}_3)}\right)^2 \sim \chi^2_{(2)}.$$

The test may be called a "Wald" test, a likelihood ratio (LR) test, or a Lagrnage multiplier (LM) test.

Generalized method-of-moments estimation

- If heteroskedasticity or serial correlation is present in a model with endogenous variables, then using IV estimation with a ``robust" covariance matrix ensures that interval estimators, hypothesis tests, and prediction intervals use a valid standard error.
- There is a GMM estimator that is “aymptotically” more efficient than the IV estimator, i.e, the GMM estimator has smaller variances than the IV estimator in large samples.

Goodness-of-fit with IV estimates

We discourage the use of R^2 outside the context of OLS estimation.

When there are endogenous variables on the right-hand side of a regression equation,

$$\text{SST} \neq \text{SSR} + \text{SSE}\,.$$

R^2 could be negative when based on IV estimates!

R

This is okay to get the 2SLS estimates, but the standard errors are wrong.

The first-stage equation

```
data("mroz", package="PoEdata")
mroz1 <- mroz[mroz$lfp==1,] #restricts sample to lfp=1
educ.ols <- lm(educ~exper+I(exper^2)+mothereduc, data=mroz1)

educHat <- fitted(educ.ols)
wage.2sls <- lm(log(wage)~educHat+exper+I(exper^2), data=mroz1)
```

The second-stage equation

R: use ivreg() instead

```
mroz1 <- mroz[mroz$lfp==1,] #restricts sample to lfp=1.
mroz1.ols <- lm(log(wage)~educ+exper+I(exper^2), data=mroz1)
mroz1.iv <- ivreg(log(wage)~educ+exper+I(exper^2)|
            exper+I(exper^2)+mothereduc, data=mroz1)
mroz1.iv1 <- ivreg(log(wage)~educ+exper+I(exper^2)|
            exper+I(exper^2)+mothereduc+fathereduc,
            data=mroz1)
```

Example: R

Table 1: Wage equation: OLS, 2SLS, and IV models compared

	Dependent variable: wage			
	OLS	explicit 2SLS	IV mothereduc	IV mothereduc and fathereduc
Constant	−0.5220 (0.1986)	0.1982 (0.4933)	0.1982 (0.4729)	0.0481 (0.4003)
educ	0.1075 (0.0141)		0.0493 (0.0374)	0.0614 (0.0314)
educHat		0.0493 (0.0391)		
exper	0.0416 (0.0132)	0.0449 (0.0142)	0.0449 (0.0136)	0.0442 (0.0134)
I(exper^2)	−0.0008 (0.0004)	−0.0009 (0.0004)	−0.0009 (0.0004)	−0.0009 (0.0004)
Observations	428	428	428	428

10.4 Specification Tests

10.4 Specification tests

Good IV

- $cov(z, e) = 0$
- $corr(z, x) \neq 0$
- z can not be put in the RHS of the structural equation

1. 10.3.9: Weak instruments test, $H_0 : z_1, \ldots, z_L$ are weak instruments.

   Regress endogenous variable x on $z_1, \ldots, z_L$ and use an F test.

2. 10.4.3: Overidentifying restrictions test (Sargen test):

H_0: instruments are valid, i.e., $cov(z_l, e) = 0$ for $l = 1, \ldots, L$.

3. 10.4.1-2: Hausman test, $H_0 : x$ is exogenous, i.e., $cov(x, e) = 0$

10.3.9 Assessing instrument strength using the first stage model

- Case 1: Assessing the strength of one IV. Suppose the first stage regression is

$$x_K = \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_{K-1} x_{K-1} + \theta_1 z_1 + \nu_K \,.$$

The key to assessing the strength of the IV z_1 is the strength of its relationship to x_k after controlling for the effects of all the other exogenous variables.

z_1 must be a statistically significant effect on x_k .

- Case 2: Assessing the strength of more than one IV: Suppose the first stage regression equation is

$$x_K = \gamma_1 + \gamma x_2 + \cdots + \gamma_{K-1} x_{K-1} + \theta_1 z_1 + \cdots + \theta_L z_L + \nu_K.$$

We require that at least one of the instruments be strong.

If the F-value is not sufficiently large, then IVs and 2SLS estimation is quite possibly worse than "ordinary" LS.

R

- In the first-stage regression model:

```
educ.ols <- lm(educ~exper+I(exper^2)+mothereduc+fathereduc,
               data=mroz1)
```

- Conduct the hypothesis test:

```
linearHypothesis(educ.ols, c("mothereduc=0", "fathereduc=0"))
```

10.4.3. Testing Instruments Validity

Recall

To test H_0: the (multiple) regression model is not useful in predicting the response variable, we employ the test statistic:

$$NR^2 \sim \chi^2_{(df)}\,.$$

This is a right-tailed test, because

- if H_0 is true, R^2 is close to zero, and NR^2 is small.
- If H_0 is false, R^2 is close to 1, and NR^2 becomes large.

Logic

Suppose H_0: IVs are valid is true.

$\iff cov(z_l, e) = 0 \, for \, l = 1, \ldots, L$

$\iff$ The following regression can not predict e:

$$e = \gamma_1 + \theta_1 z_1 + \cdots + \theta_L z_L + v \, .$$

$\iff H_0$: the regression model in (1) is not useful in predicting e

.

Procedures

1. Compute the IV estimates $\hat{\beta}_k$ using all available IV, including the G exogenous variables and L instruments, $z_1, \ldots, z_L$.
2. Obtaintheresiduals: $\hat{e}_{IV} = y - \hat{\beta}_1 - \hat{\beta}_2 x_2 - \cdots - \hat{\beta}_K x_K$.
3. Consider the regression: $\hat{e}_{IV} = \theta + \gamma_1 z_1 + \cdots + \gamma_L z_L$.
4. Compute NR^2 from the above regression.
5. If all of the surplus moment conditions are valid, the test statistic is $NR^2 \sim \chi^2_{(L-B)}$, and this is a right-tailed test.

10.4.1 The Hausman test for endogeneity

- $$\begin{cases} H_0 : cov(x, e) = 0 & \text{(exdogenous)} \\ H_1 : cov(x, e) \neq 0 & \text{(endogenous)} \end{cases}$$
- If H_0 is true, both LS and IV estimators are consistent. Then, we use the more efficient LS estimator.
- If H_0 is false, the LS estimator is not consistent. We use the consistent IV estimator.

Hausman test for endogeneity 1/3: $B = 1$ and $L = 1$

Consider the model: $y = \beta_1 + \beta_2 x + e$. Let z be IVs for x.

1. Estimate the model $x = \gamma_1 + \theta_1 z + \nu$ by LS to obtain the residuals $\hat{\nu} = x - \hat{\gamma}_1 - \hat{\theta}_1 z$.
2. Consider the model:

$$y = \beta_1 + \beta_2 x + \delta \hat{\nu} + e \, .$$

3. Estimate the above "artificial regression" by LS and use the t test for

$$H_0 : \delta = 0$$

$$H_1 : \delta \neq 0$$

Why?

10.4.2 The logic of the Hausman Test: $B = L = 1$

Consider

$$y = \beta_1 + \beta_2 x + e, \qquad (10.32)$$

Suppose z is a good IV.

$$x = \gamma_1 + \theta_1 z + v = E(x \mid z) + v,$$

LS estimation gives:

$$\hat{x} = \hat{\gamma}_1 + \hat{\theta}_1 z,$$

We set

$$\hat{v} = x - \hat{x},$$
$$x = \hat{x} + \hat{v}\,.$$

10.4.2 The logic of the Hausman Test: $B = L = 1$

Estimate (10.32) with LS.

$$y = \beta_1 + \beta_2 x + e \qquad (10.32)$$
$$= \beta_1 + \beta_2(\hat{x} + \hat{v}) + e$$
$$= \beta_1 + \beta_2\hat{x} + \gamma\hat{v} + e \qquad (10.38)$$

Add and subtract $\beta_2\hat{v}$:

$$y = \beta_1 + \beta_2\hat{x} + \gamma\hat{v} + e + \beta_2\hat{v} - \beta_2\hat{v}$$
$$= \beta_1 + \beta_2(\hat{x} + \hat{v}) + (\gamma - \beta_2)\hat{v} + e$$
$$= \beta_1 + \beta_2 x + \delta\hat{v} + e \qquad (10.39)$$

Suppse "H_0 : x is exogenous" is true

$\iff$ x is exogenous

$\iff$ $\hat{\gamma}_{LS} \to^p \beta_2$ in (10.38)

$\iff$ $\gamma = \beta_2$ in (10.38)

$\iff$ $\delta = 0$ in (10.39)

Suppose "H_0: x is exogenous" is false

$\iff$ x is endogenous

$\iff$ $\hat{\gamma}_{LS}$ does not converge to β_2 in probability in (10.38)

$\iff$ $\gamma \neq \beta_2$ in (10.38)

$\iff$ $\delta \neq 0$ in (10.39)

Hausman test for endogeneity 2/3: $B = 1$ and $L = 2$

Consider the model: $y = \beta_1 + \beta_2 x + e$. Let z_1, and z_2, be IVs for x.

1. Estimate the model $x = \gamma_1 + \theta_1 z_1 + \theta_2 z_2 + \nu$ by LS, and obtain the residuals $\hat{\nu} = x - \hat{\gamma}_1 - \hat{\theta}_1 z_1 - \hat{\theta}_2 z_2$.
2. Consider the model:

$$y = \beta_1 + \beta_2 x + \delta \hat{\nu} + e \ .$$

3. Estimate the above "artificial regression" by LS and use the t test for

$$H_0 : \delta = 0$$

$$H_1 : \delta \neq 0$$

Hausman test for endogeneity 3/3: $B > 1$ and $L \geq 1$

1. Repeat step 1 for each one, using all IVs in each regression.
2. Replace the model in step 2:

$$y = \beta_1 + \beta_2 x_2 + \cdots + \beta_{B+1} x_{B+1} + \delta_1 \hat{\nu}_1 + \cdots + \delta_B \hat{\nu}_B + e \, .$$

3. Estimate the above "artificial regression" by LS and use the F test for

$$H_0 : \delta_1 = \cdots = \delta_B = 0$$

$$H_1 : \textit{ at least one } \delta_i \textit{ in } H_0 \textit{ is not zero.}$$

R and Interpretations

- Weak instruments test: rejects the null, meaning that at least one instrument is strong
- (Wu-)Hausman test for endogeneity: barely rejects the null that the variable of concern is uncorrelated with the error term, indicating that educ is marginally endogenous
- Sargan overidentifying restrictions: does not reject the null, meaning that the extra instruments are valid (are uncorrelated with the error term)

R automatically performs these three tests:

```
summary(mroz1.iv1, diagnostics=TRUE)
```

Outputs

```
## Diagnostic tests:
##                    df1 df2 statistic p-value
## Weak instruments     2 423     55.40  <2e-16 ***
## Wu-Hausman           1 423      2.79   0.095 .
## Sargan               1  NA      0.38   0.539
```

For R demo, see

https://bookdown.org/ccolonescu/RPoE4/random- regressors.html

End of this Chapter 10!